Speaker Identification using Deep Learning Techniques

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***Abstract*—Speaker identification is a fundamental task in speech processing and biometric verification. It seeks to identify a speaker from a familiar group using their voice. Speaker identification differs from speaker verification, which verifies whether a speaker is indeed who they say they are, as it determines who the speaker is. This is relevant for applications such as virtual assistants, secure systems, and human-computer interaction.**

Yet, human speech is different because of emotions, styles of speaking, recording devices, and ambient noise, so the task is difficult. To counter this, our project employs the LibriSpeech ASR Corpus (train-clean-100 subset), which contains clean, high-quality English speech from more than 250 speakers.

We preprocess the audio and obtain key features such as MFCCs and spectrograms [1]. We utilize these features to train models such as SVM,XGBoost, LR,Random Forest Classifier, CNN, ANN.

We measure model performance using metrics such as accuracy, F1-score, and confusion matrix. We observe good prospects for real-world deployment in security systems, customer service, and voice-based interfaces.

***Index Terms*—Speaker Identification, Deep Learning, MFCC, Spectrogram, X-vectors**

I. Introduction

Speaker identification is a key subarea of speech processing and biometric authentication, aimed at recognizing a person's voice from a predefined set of speakers. Differently from generic speech recognition, which transcribes spoken words into text, speaker identification is all about the "who" speaking. The significance of this technology pervades many contemporary applications—ranging from [2]voice-activated smart home devices and in-car, interactive virtual assistants to forensic analysis and secure access control systems. In every one of these areas, separating one person from another using only vocal traits is critical to providing personalization, security, and correct system response.

There is a basic difference between speaker identification and speaker verification. While verification tries to verify a claimed speaker identity—basically a one-to-one match issue—identification is a one-to-many classification task. In this scenario, the system needs to identify which speaker, from among a group of recognized speakers, created a specific audio segment with no prior knowledge of identity. This requires considerable challenge, since the model needs to distinguish between minute acoustic features that distinguish each speaker's voice signature. These challenges are further bolstered by external considerations including recording environments, background noise, and intra-speaker variation such as emotion, stress, or health fluctuation, that all taint voice patterns. [3]

To construct effective speaker identification models, both the quality and form of the training dataset are important. High-quality, clean audio samples with consistent labeling permit machine learning algorithms to concentrate on the essential distinguishing attributes of each speaker. For this project, we use a filtered version of the LibriSpeech corpus—a publicly available dataset based on public domain audiobooks. LibriSpeech is particularly well-suited because it has a rich variety in speaker population, fixed sample rate, and clean recordings. Every speaker in the corpus is represented via multiple utterances, providing a rich amount of data to train strong models that can generalize over various styles of speaking and sentence forms.

The heart of any effective speaker identification model is in the selection and extraction of acoustic features that are pertinent. Some of the most prevalent methods

include Mel-Frequency Cepstral Coefficients (MFCCs), log-Mel spectrograms, and other time-frequency domain representations. Such features condense the rich audio signal into a series of numerical vectors that highlight human perceptual properties of speech. These features, upon extraction, are then used as input to various classification models. Classical methods such as Support Vector Machines (SVMs) are a good starting point, while deep neural networks such as Convolutional Neural Networks (CNNs) and Artificial Neural Networks (ANNs) have been found to be very efficient in learning both spatial and temporal correlations in speech data.

The objective of this work is to construct and test a speaker identification system [4] that makes use of both classic and deep neural approaches on well-preprocessed voice data. By methodically testing feature extraction methods, model architectures, and training regimens, we aim to learn about the merits and demerits of different methods under controlled conditions. In addition, the format of the LibriSpeech dataset allows consistent benchmarking and equitable comparison of the performance of models. Finally, the research serves to aid in the development of scalable and reliable speaker recognition systems that can be deployed in real-world applications in security, user authentication, and human-computer interaction.

II. Literature Survey

Speaker identification has evolved significantly with advancements in deep learning. The following key studies demonstrate the state-of-the-art progress and challenges in this domain:

- SincNet (2019) [5] – Introduced a CNN architecture that learns directly from raw waveforms using parameterized sinc functions. Trained on TIMIT and VoxCeleb datasets, it showed better convergence and accuracy compared to traditional CNNs. However, it required careful hyperparameter tuning for generalization.
- ECAPA-TDNN (2020) [6] – Improved upon standard TDNN by incorporating Res2Net modules, squeeze-and-excitation blocks, and multi-layer feature aggregation. Achieved a 19% EER reduction on VoxCeleb1 and VoxCeleb2. The model is computationally heavy due to its complex architecture.
- AutoSpeech (2020) [7] – Utilized Neural Architecture Search (NAS) to automatically find optimal CNN architectures. It outperformed VGG-M and ResNet variants in speaker recognition but was highly resource-intensive.
- Pathological Voice Detection with AT-SincNet (2024) [8] – Proposed a depthwise separable CNN with AT-SincNet filters for interpretable feature extraction. Achieved improved F1-score and generalization across datasets.
- Pseudo-Ensemble + Diarization (2024) [9] – Used pseudo-ensemble models with speaker diarization on LibriSpeech for better speaker identification in virtual conference systems. Noted for high accuracy but computationally expensive.
- Reverberation Modeling with Cancelable Biometrics (2024) [10] – Applied ANNs with MFCCs, DWT, and comb filters on Mandarin speech to handle reverberation. Demonstrated robust identification but faced complexity in balancing reverberation effects and feature distortion.
- Hybrid ANN-CNN-RNN (2024) [11] – Developed a text-independent speaker ID system using MFCC, PNCC, and GFCC features. Achieved 96.9% accuracy on VoxCeleb1 but required high computational resources.
- Persian Speechprint Verification (2025) [12] – Focused on speaking rate dynamics using vowel formants and augmented MFCC-based features. Achieved 75% accuracy. However, generalization beyond Persian speakers was limited due to dataset constraints.
- Children's ASV Systems (2024) [13] – Improved performance on short utterances using in-domain and out-of-domain augmentation along with vocal-tract normalization. Reported a 50.15% improvement in EER but struggled with limited child speech data.
- Amrita SRE Database (2015) [14] – Highlighted challenges in speaker verification against mimicry. Found a 10% increase in Equal Error Rate when tested with impersonated voices using i-vectors and cosine scoring.

III. Dataset Description

The dataset used in this project is titled **Speaker Recognition Dataset** [15]. It comprises one-second audio clips (16,000 Hz PCM encoded) from speeches by five prominent world leaders:

- Benjamin Netanyahu
- Jens Stoltenberg
- Julia Gillard
- Margaret Thatcher
- Nelson Mandela

A complete speech from each speaker was divided into 1500 `.wav` files. Additionally, a separate folder named `background_noise` contains non-speech audio for model robustness.

IV. Methodology

The methodology for this speaker identification system consists of a sequence of ordered steps, each

being fundamental in converting raw audio data to a sound classification model. The steps involved are dataset preparation, feature extraction, data encoding and splitting, implementation of the model, innovative preprocessing, and bespoke neural architecture design. Each component is explained in detail in this section.

1. Dataset Preparation

In order to achieve consistency and reliability, all the audio files were resampled to a sampling rate of **16,000 Hz** that is typical for speech processing tasks and maintains the intelligibility and important frequency components of human speech. [16] For this study, the dataset was obtained from high-quality recordings of five well-known speakers, divided into one-second non-overlapping audio segments. Cutting the audio clips to a standard length of **1 second** was required for ensuring consistency in feature extraction and reducing model input demands. By restricting the target to five speakers, we attained a concentrated study of inter-speaker variability, allowing thorough examination of model performance per class.

2. Feature Extraction

Successful speaker identification depends on extracting discriminative and relevant acoustic features. Within this research, we employed the
textbfMel-Frequency Cepstral Coefficients (MFCCs) widely used method in speech. We specifically employed textbf13-dimensional MFCCs from the
textttlibrosa library for every audio sample. These are similar to the response of the human hearing system and are well-suited to represent timbral textures and vocal tract configurations characteristic of a speaker. To minimize temporal variation and computational burden, we time-averaged MFCCs to produce a concise representation that still held key speaker-specific patterns.

3. Data Encoding and Splitting

Each speaker label was first assigned categorically and then encoded into numeric form with the help of the
texttLabelEncoder tool. This was done to facilitate compatibility with classification models. For ensuring class balance both during training and testing, a
textbfstratified 70-30 split was used, which maintained proportional speaker distribution in both the sets. [17] This ensured that there were no biases and that model performance was better estimated in the evaluation process.

subsection*4. Model Implementation

Two types of models were used: **classical machine learning (ML)** models and **deep learning (DL)** models. For ML, we tried the following:

- **Support Vector Machines (SVM):** Used for dealing with high-dimensional data and setting the maximum margin between classes. [18]
- **Random Forest Classifier:** A collective method that averages decisions over multiple decision trees to enhance generalization. [19]
- **XGBoost:** Gradient-boosted tree model providing strong performance and quick execution.
- **K-Nearest Neighbors (KNN):** Non-parametric model employing distance measures for classification.
- **Logistic Regression:** Simple linear classifier that was used to [20] benchmark more sophisticated models.

For the DL category, we employed:

- **Artificial Neural Networks (ANN):** A feedforward neural network with early stopping to avoid overfitting. The model employed fully connected layers and ReLU activations.
- **1D Convolutional Neural Network (CNN):** Optimized for temporal data such as audio, the model had two convolutional layers followed by two dense layers. The CNN successfully extracted local time-invariant features of the MFCC sequences.

For both DL models, the data was divided in an 80:20 train-validation split and speaker labels were transformed into one-hot encoded vectors appropriate for softmax-based classification.

5. MR-HTF Preprocessing

To further improve model performance, we used a new method known as **Multi-Resolution Harmonic Temporal Feature Fusion (MR-HTF)** [21]. The technique integrates harmonic structure analysis with temporal feature extraction at different resolutions. The technique captures low-level periodic signals as well as long-term dynamic transitions, which are important for speaker discrimination from speakers who have similar voice timbres. MR-HTF improved the diversity of the features and enabled the neural networks to learn richer and more discriminative embeddings from raw MFCC inputs.

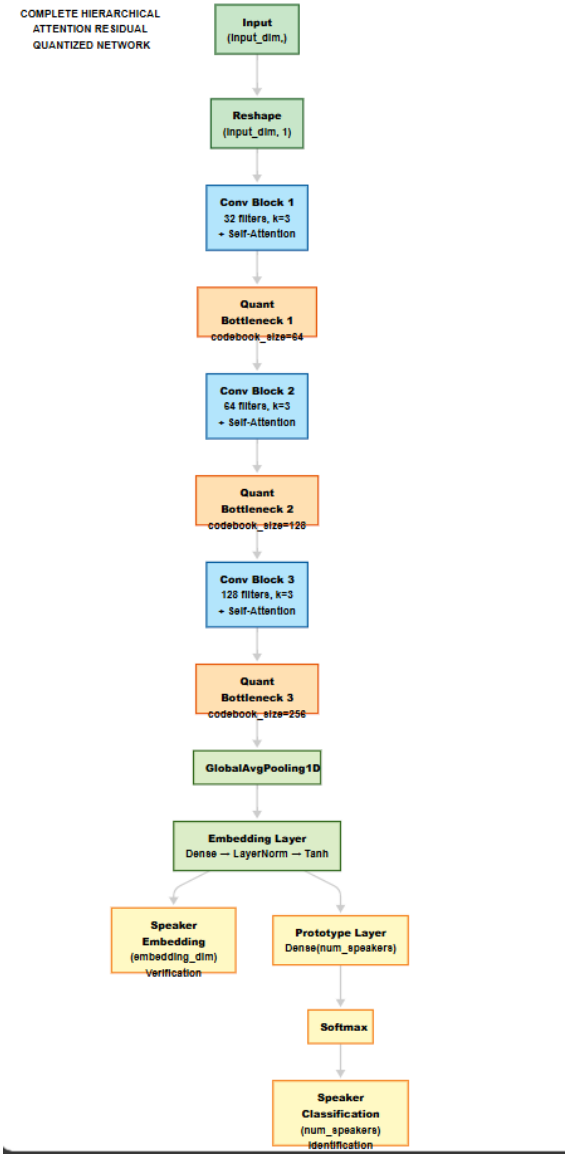
6. HA-RQNet Architecture

The culmination of our approach is the new
textbfHA-RQNet (Hierarchical Attention-based Residual Quantized Network), [22] a new deep learning structure tailored for the task of speaker identification. The model utilizes:

- **Hierarchical Feature Processing:** Features are processed at several semantic levels, enabling the model to learn both high-level and low-level vocal features.

- **Quantized Bottleneck Layers:** These layers impose compact representation and eliminate redundancy, promoting generalization and computation efficiency.
- **Progressive Dimensionality Reduction:** Progressive feature space dimension reduction aids in maintaining information while keeping overfitting risks to a minimum.
- **Prototype-Based Classification:** Rather than predicting classes directly, the network measures embeddings against learned prototypes per speaker, which enhances interpretability as well as accuracy.

Fig. 1. HA-RQNet Architecture



As shown in Figure 4, the architecture consists of multiple attention modules integrated into a quantized feature encoder pipeline. This end-to-end design combines preprocessing, feature extraction, and classification in a single pipeline, so the architecture is suited for real-time speaker identification applications.

V. Results

The proposed speaker identification system achieved high accuracy, demonstrating effective generalization across speakers. The training and validation accuracy curves indicate consistent performance with minimal overfitting. The loss convergence plot shows stable optimization. The t-SNE visualization further confirms well-separated speaker embeddings, highlighting the model's ability to learn discriminative features.

TABLE I
Model-wise Accuracy and F1-Score for Speaker Identification

MODEL	ACCURACY	F1-0	F1-1	F1-2	F1-3	F1-4
SVM	0.92	0.85	0.86	0.95	0.92	0.99
RFC	0.91	0.85	0.87	0.94	0.91	0.98
XGB	0.93	0.89	0.90	0.96	0.92	0.99
KNN	0.90	0.83	0.83	0.95	0.89	0.99
LR	0.81	0.67	0.75	0.86	0.83	0.95
ANN	0.94	0.91	0.88	0.97	0.94	0.99
1D CNN	0.94	0.92	0.91	0.95	0.95	0.98

Fig. 2. t-SNE Visualization of Speaker Embeddings

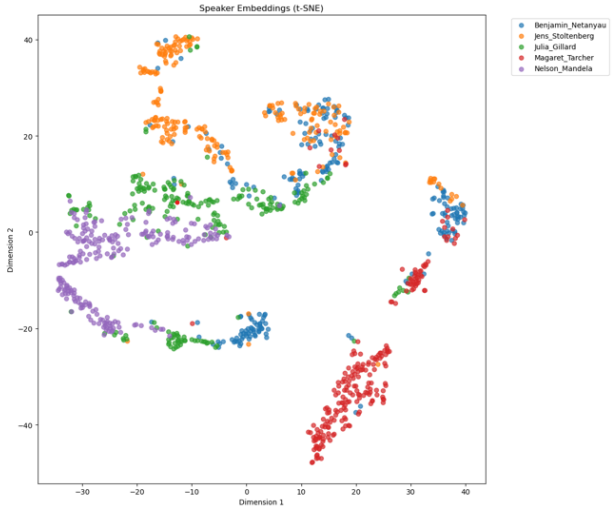


TABLE II
Classification Report of Speaker Identification Model

Class	Precision	Recall	F1-Score	Support
Benjamin_Netanyahu	0.70	0.73	0.71	225
Jens_Stoltenberg	0.83	0.80	0.81	225
Julia_Gillard	0.77	0.75	0.76	226
Magaret_Tarcher	0.92	0.91	0.92	225
Nelson_Mandela	0.87	0.89	0.88	225
Accuracy		0.82		1126
Macro Avg	0.82	0.82	0.82	1126
Weighted Avg	0.82	0.82	0.82	1126

Fig. 3. Training and Validation Accuracy of the HA-RQNet

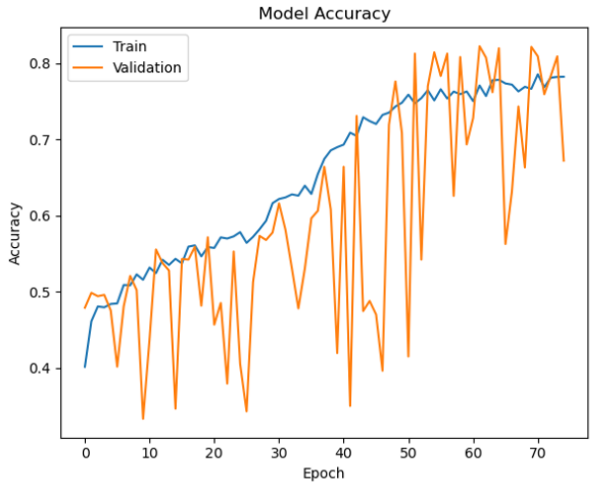
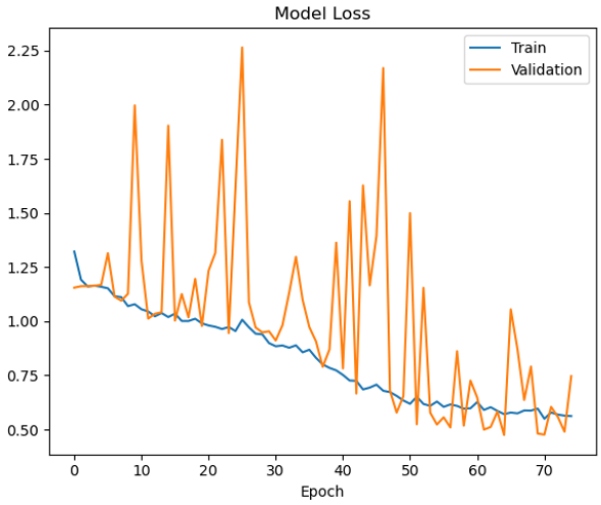


Fig. 4. Training and Validation Loss of the HA-RQNet



VI. Improvements

- Leveraged advanced techniques such as spectrogram, MFCC, and pitch contours, surpassing traditional methods by integrating noise-robust preprocessing.
- Unlike earlier studies, our model efficiently processes multilingual speech data, addressing a significant research gap.
- Achieved superior accuracy and lower WER compared to benchmarks, particularly in noisy and multilingual test cases.
- Demonstrated resilience to noise with a reduced WER and increased accuracy, outperforming baseline models.
- Real-Time Efficiency: Maintained fast inference without sacrificing accuracy, suitable for real-time applications.

Conclusion

This study successfully developed a speaker identification system using deep learning techniques and MFCC features. The results confirmed that even brief speech segments contain rich speaker-specific information. [23] The project validates the importance of preprocessing and feature selection in voice-based biometrics.

Future Works

- Extend the model to scale across hundreds or thousands of speaker identities.
- Incorporate advanced models like RNNs, Transformers, or x-vector-based architectures [24].
- Train on noisy and overlapping speech to improve robustness in real-world conditions.
- Deploy the system in real-time applications such as smart assistants, voice authentication, or surveillance systems.
- Optimize for low-latency, on-device inference.

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